

Factorial:

$$n! = n \times (n - 1) \times (n - 2) \times \dots \times 2 \times 1$$

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

$$0! = 1$$

Permutations:

Objects are **arranged** in a specific **order**

How many ways are there of arranging the letters A, B and C in a straight line?

A B C

A C B

B A C

B C A

C B A

C A B

= 6

$$3! = 6$$

The number of ways of arranging n objects in a straight line are $n!$

How many ways are there of arranging 1 letter from A, B and C?

A

B

C

= 3

$${}^3P_1 = 3$$

$${}_rP = \frac{n!}{(n-r)!}$$

Combinations:

Objects are **selected**

How many ways are there to select 5 balls from a box of 20?

$${}_{20}C_5 = 15504$$

$${}_nC_r = \frac{n!}{(n-r)!r!}$$